nanofem

A 3D electromagnetic field solver in 1000 lines of Rust

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Abstract

nanofem solves the time harmonic curl-curl equation for the electric field on a tetrahedral mesh with first order Nedelec edge elements, and reports scattering parameters at lumped ports. It covers perfect conductors, a first order absorbing boundary, perfectly matched layers, dielectric, conductive and magnetic materials, lossy conductor sheets with the skin effect, and field export. The system is complex symmetric and is factored directly after a nested dissection ordering, with symmetric equilibration and one step of iterative refinement. The entire solver lives in one Rust source file of at most 1000 nonblank, noncomment lines, enforced by a test, with no dependencies outside the standard library. This report derives each algorithm, walks through the file in order, and explains the design decisions the budget forced. The bar under each section heading shows where that code section sits in the line budget.

1 The budget as a design tool

The rule: one file, `src/main.rs`, at most 1000 lines after removing blank lines and comments, standard library only. A test reads the file, counts, and fails the build above the limit. Tests, meshes and comments are free. The budget forces every feature to displace another, which makes the cost of each decision explicit, and it keeps the solver readable in one sitting. That is the purpose of the repository.

This report follows the source file from top to bottom. Every section corresponds to a part of the file and carries a bar showing which slice of the budget it occupies. Table 1 is the overview.

code section	lines	cumulative
constants, error and output plumbing	13	13
complex arithmetic	36	49
vectors, tetrahedron geometry, Whitney mass	22	71
Gmsh mesh reader	87	158
deck parser	88	246
nested dissection ordering	56	302
sparse LDL^T, equilibration, refinement	157	459
element tables and port data	8	467
output rows and parameter conversion	51	518
roles, materials, PML tensors	73	591
edges, unknowns, group echo	38	629
port geometry	22	651
volume and boundary assembly	76	727
matrix build and symbolic analysis	46	773
frequency sweep and scattering parameters	110	883
field export	62	945
main	9	954

Table 1: Line budget by code section, in file order.

The linear algebra is the largest block, at 213 lines for the ordering and the factorization. The input format is small in comparison, which is the reverse of the distribution in nanospice. Everything the budget excluded is listed with reasons in section 11.

2 The formulation

`theory only` 0 lines

Maxwell's equations with an $e^{+j\omega t}$ time dependence give, after eliminating the magnetic field,

$$\nabla \times \left(\frac{1}{\mu_r} \nabla \times \mathbf{E} \right) - k_0^2 \varepsilon_r \mathbf{E} = -jk_0 \eta_0 \mathbf{J}, \qquad k_0 = \frac{\omega}{c_0}, \tag{1}$$

with η_0 the free space wave impedance. This is the only equation the solver discretizes. Multiplying by a test function $\mathbf{W}$ and integrating by parts over the domain Ω yields the weak form

$$\int_\Omega \frac{1}{\mu_r} (\nabla \times \mathbf{W}) \cdot (\nabla \times \mathbf{E}) - k_0^2 \int_\Omega \varepsilon_r \mathbf{W} \cdot \mathbf{E} + \oint_{\partial\Omega} \mathbf{W} \cdot (\hat{n} \times \nabla \times \mathbf{E}) = -jk_0 \eta_0 \int_\Omega \mathbf{W} \cdot \mathbf{J}. \tag{2}$$

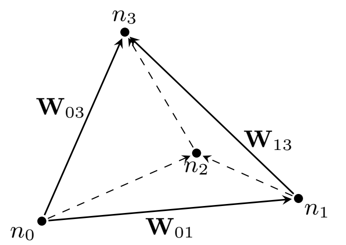
The three volume and surface terms map one to one onto the three matrices the code assembles: a curl stiffness, an ε weighted mass, and a boundary term that carries every absorbing and port condition. Dropping the boundary integral leaves the natural boundary condition of the formulation, which is a perfect magnetic wall. A magnetic symmetry plane therefore needs no card in the deck.

Design decision: edge elements rather than nodal elements
Expanding **E** in nodal basis functions, one scalar unknown per component per node, fails for this equation in two ways. It cannot represent the field discontinuity normal to a material interface, and it produces spurious modes: the discrete curl-curl operator acquires eigenvectors with no physical counterpart, which contaminate resonances and scattering alike. Whitney edge elements assign one unknown to the tangential field along each edge. Tangential continuity across faces is then automatic and normal discontinuity is allowed, which is exactly the physical regularity of **E**. The gradient fields sit exactly in the kernel of the discrete curl, so they are represented rather than approximated, and no spurious modes appear. The cost is six unknowns per tetrahedron instead of twelve.

The Whitney function of the edge from local vertex a to local vertex b is built from the barycentric coordinates λ of the tetrahedron,

$$\mathbf{W}_{ab} = \lambda_a \nabla\lambda_b - \lambda_b \nabla\lambda_a, \qquad \nabla \times \mathbf{W}_{ab} = 2\,\nabla\lambda_a \times \nabla\lambda_b. \tag{3}$$

Its tangential component along its own edge is one and along the other five edges is zero, so the unknown attached to an edge is the line integral of the field along it. The curl is constant over the element.



edge	nodes
0	(n_0, n_1)
1	(n_0, n_2)
2	(n_0, n_3)
3	(n_1, n_2)
4	(n_1, n_3)
5	(n_2, n_3)

Figure 1: The six edge unknowns of a tetrahedron and the local edge table the code uses. The unknown on an edge is the line integral of **E** along it. Node indices are sorted on input, so every arrow runs from the lower to the higher global index and two tetrahedra sharing an edge assign it the same direction without a sign convention. Edges meeting the rear node n_2 are dashed.

Derivation: the element matrices in closed form

The gradients $\nabla\lambda$ are constant over a straight tetrahedron, so the stiffness entry follows immediately from (3),

$$S_{ij} = \int_T \frac{1}{\mu_r}(\nabla \times \mathbf{W}_i) \cdot (\nabla \times \mathbf{W}_j) = \frac{4V}{\mu_r}(\nabla\lambda_a \times \nabla\lambda_b) \cdot (\nabla\lambda_c \times \nabla\lambda_d),$$

with V the element volume and (a, b), (c, d) the vertex pairs of edges i and j. The mass entry needs only the integrals of products of barycentric coordinates, which are $\int_T \lambda_p \lambda_q = V/10$ for $p = q$ and $V/20$ otherwise, giving the four term expression

$$M_{ij} = (\nabla\lambda_b \cdot \nabla\lambda_d)I_{ac} - (\nabla\lambda_b \cdot \nabla\lambda_c)I_{ad} - (\nabla\lambda_a \cdot \nabla\lambda_d)I_{bc} + (\nabla\lambda_a \cdot \nabla\lambda_c)I_{bd}.$$

No quadrature is needed anywhere in the element loop. The same expression, with the surface gradients of a triangle and $I_{pq} = A/6$ or $A/12$, gives the face mass matrix used by every boundary condition, so one helper serves both.

3 Dispersion and mesh density

`theory and measurement`	0 lines

The discrete dispersion error of an edge element of order p scales as $(kh)^{2p}$, so the phase a wave accumulates over a length L carries a relative error that grows as k^{2p+1}. For the first order elements used here this means the phase error rises with the cube of frequency at fixed mesh, and the mesh must be sized by the shortest wavelength in the model. Table 2 gives the measured error over the reference line and fixes the mesh density: ten cells per wavelength give a relative phase error near one percent, thirty give one part in fifteen hundred.

cells per wavelength	phase error / rad	relative
30.0	1.63×10^{-3}	6.48×10^{-4}
10.0	4.53×10^{-2}	6.00×10^{-3}
6.0	2.22×10^{-1}	1.77×10^{-2}
3.7	1.04×10^{0}	5.17×10^{-2}

Table 2: Accumulated phase error of a TEM line against its exact value $-k_0L$. The measured growth per factor three in frequency is 27.8, against 27 from the k^3 law.

Failure mode: an under-resolved mesh shifts a resonance without failing

Dispersion error slows the discrete wave, which lengthens the electrical size of a structure and moves every resonance down. Nothing in the output marks this: the scattering parameters remain passive, reciprocal and smooth, the pivot spread and the residual stay small, and the only symptom is that the answer is wrong. The reference antenna in section 9 resonates at 2.48 GHz on a mesh graded from 3 to 9 mm and at 2.50 GHz on the mesh of section 9, refined by a factor of one and a half. Mesh convergence has to be established by running two densities, since no diagnostic in the solver can substitute for it.

4 The mesh reader and orientation

`Gmsh mesh reader`	87 lines, budget 71 to 158

The Whitney function of an edge depends on which of its two vertices is called a. Two tetrahedra sharing an edge must agree, or their contributions cancel instead of adding. The textbook remedy is to give every global edge a reference direction and to carry a sign through assembly, excitation and field evaluation.

Design decision: sort the element vertices when reading the mesh

The mesh reader sorts the four node indices of every tetrahedron and the three of every triangle into ascending order. Every local edge (a, b) then satisfies $a < b$ in global numbering, so all elements agree on the direction of every shared edge by construction. Assembly, the port excitation integral and the field export are then all sign free. One line in the reader replaces about ten sign carrying expressions elsewhere, and a sign error in any of them cancels contributions instead of adding them, which changes the result without failing visibly.

Sorting changes the sign of the Jacobian determinant, which is why the volume is taken as $|\det|/6$; the barycentric gradients themselves are geometric quantities and do not depend on the vertex order.

5 Materials and the perfectly matched layer

`roles, materials, PML tensors` 73 lines, budget 518 to 591

A material is a relative permittivity with a loss tangent, a relative permeability, and a conductivity. A perfectly matched layer is not a material but a coordinate transformation: stretching coordinate k into the complex plane by $s_k = 1 - ja_k$ turns an outgoing wave into a decaying one without reflecting it. Writing the stretched operator back in ordinary coordinates turns the stretch into a diagonal tensor applied to both ε and μ,

$$\Lambda_k = \frac{s_x s_y s_z}{s_k^2}, \qquad \varepsilon \to \varepsilon\Lambda, \qquad \mu \to \mu\Lambda. \tag{4}$$

Applying the same tensor to both keeps the wave impedance of the layer equal to that of the region it terminates, which is what makes it reflectionless at the continuous level. The code stores Λ alone and multiplies the resulting mass entry by ε, which costs one tensor contraction per element pair instead of two and is what lets conductivity reuse the same entry.

A first order absorbing boundary is the cheaper alternative: the surface term of (2) is replaced by $jk_0\sqrt{\varepsilon/\mu}$ times the tangential face mass matrix, which is exact for a plane wave at normal incidence and degrades with angle. It needs no extra mesh, while a PML needs a layer of elements. With three cells the PML reflects at -30.7 dB in the test suite, and a stretch of the same magnitude on all three axes instead of one costs less than a decibel of that, so a single shell region serves the faces, edges and corners of a box alike. Against the extra elements, an absorbing boundary has to stand off a quarter wavelength or more from a radiator to be accurate at the angles that matter, while a PML does not: the reference antenna of section 9 needs a third of the air height it needed with an absorbing boundary.

6 Loss and the frequency decomposition

`volume and boundary assembly` 76 lines, budget 651 to 727

Every entry of the system matrix is a sum of terms with a known dependence on k_0. The curl stiffness is constant, the mass enters as $-k_0^2$, and the boundary terms as $+jk_0$. Without loss the matrix is a quadratic polynomial in k_0, so the assembly can run once for an entire sweep. Two of the three loss mechanisms preserve that form and the third extends it by one term.

Derivation: conductivity is a term linear in the wave number

A conductive region has $\varepsilon_c = \varepsilon_r - j\sigma/(\omega\varepsilon_0)$. Substituting into the mass term of (2),

$$-k_0^2\varepsilon_c = -k_0^2\varepsilon_r + j\frac{k_0^2\sigma}{\omega\varepsilon_0} = -k_0^2\varepsilon_r + jk_0\eta_0\sigma,$$

using $\omega = k_0c_0$ and $\eta_0 = 1/(c_0\varepsilon_0)$. The conduction current is therefore a term linear in k_0 built from the same mass matrix, so it needs no new element integral, only a second scaling of an entry that already exists.

Derivation: the skin effect is a term in the square root of the wave number

A good conductor is modelled as a surface impedance $Z_s = (1+j)R_s$ with $R_s = \sqrt{\omega\mu_0/2\sigma}$. Its contribution to the boundary term is $jk_0\eta_0/Z_s$ times the face mass matrix. With $\omega\mu_0 = k_0\eta_0$,

$$\frac{jk_0\eta_0}{(1+j)\sqrt{k_0\eta_0/2\sigma}} = \sqrt{k_0}\,\frac{j}{1+j}\sqrt{2\sigma\eta_0} = \sqrt{k_0}\,\frac{1+j}{2}\sqrt{2\sigma\eta_0}.$$

The frequency dependence collapses to $\sqrt{k_0}$ with a constant coefficient.

Design decision: four coefficients per matrix entry

Each assembled entry stores four complex numbers, the coefficients against the basis $\{1,\, k_0,\, k_0^2,\, \sqrt{k_0}\}$. Evaluating a frequency is then four multiplications per nonzero. The assembly runs once regardless of how many frequencies the sweep contains. The cost is three additional complex numbers per nonzero, against a rebuild of the element loop per frequency.

7 Ordering and the direct solve

`ordering and sparse` LDL^T — 213 lines, budget 246 to 459

The system is complex symmetric, not Hermitian: the absorbing boundary and the port sheets contribute j times a real symmetric matrix. It is therefore factored as LDL^T without conjugation, up-looking, with the elimination tree and exact column counts computed once in a symbolic pass and reused for every frequency of the sweep.

Design decision: a direct solver, not an iterative one

Iterative solvers are the usual choice at this problem size. The gradient nullspace of the curl-curl operator makes Krylov methods with a diagonal preconditioner converge slowly, and the preconditioners that work on the edge space, auxiliary space methods and algebraic multigrid, are larger than the entire budget. A direct factorization has a predictable cost and no convergence parameters. The ceiling is memory, at a few tens of thousands of unknowns.

Fill is controlled by a geometric nested dissection ordering: split the unknowns at the median of the longest bounding box axis, take as separator those vertices of the second half with a neighbour in the first, order both halves recursively and the separator last. On the reference antenna this gives 1.88 million nonzeros in L against 19.1 million in the natural order, a factor of ten.

7.1 Equilibration and refinement

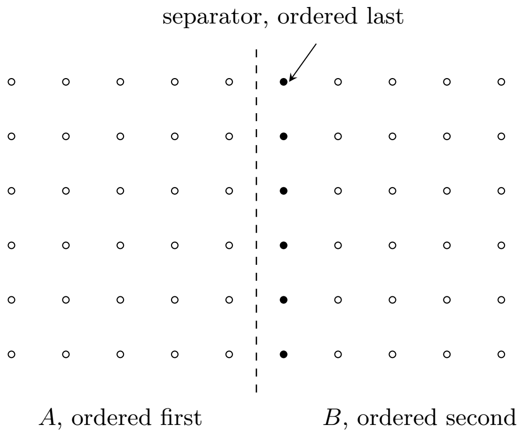


Figure 2: One level of the geometric nested dissection. The unknowns are split at the median of the longest bounding box axis. Those on the second side with a neighbour on the first form the separator and are numbered after both halves, so no fill can connect A to B. Both halves are then split recursively down to blocks of 32.

7.2 Equilibration and refinement

Without pivoting an LDL^T can lose accuracy on an indefinite matrix, and the curl-curl system is indefinite by construction above the first resonance. Bunch-Kaufman pivoting is the standard remedy and does not fit the budget. Two cheaper measures are used instead.

The matrix is first equilibrated symmetrically, DAD with $D_{ii} = |A_{ii}|^{-1/2}$, which puts the rows on a common scale. Figure 3 shows the effect on the spread of the pivots, which the factorization produces for free and which bounds the condition number from below.

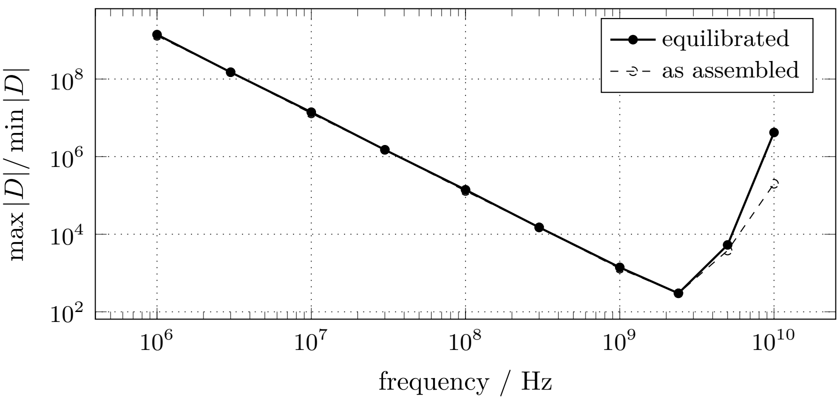


Figure 3: Pivot spread of the reference antenna against frequency. The $1/f^2$ branch on the left is the low frequency breakdown of the curl-curl operator, which loses the mass term that regularizes its nullspace. The rise on the right sets in once the mesh becomes coarse against the wavelength. Equilibration gains a factor of about three over the low frequency branch and about four in the operating band.

Every solve is then followed by one step of iterative refinement against the unscaled matrix: form $r = b - Ax$, solve $A\delta = r$ with the factorization already in hand, and add. It costs two sparse matrix vector products and one extra triangular solve, against a factorization that dominates the runtime by two orders of magnitude, and it lowers the relative residual by about a factor of four. The residual is then measured rather than assumed; it and the pivot spread are printed at the end of a run.

Design decision: report the residual instead of trusting the factorization

A direct solver returns a vector and no measure of how well it solved. Refinement already forms the residual, so reporting the worst relative residual over the sweep costs one further matrix vector product per solve. Conditioning then does not have to be inferred from the mesh and the frequency.

8 Ports and scattering parameters

`port geometry, sweep and parameters` 132 lines, budget 629 to 651 and 773 to 883

A lumped port is a rectangular patch of boundary carrying a sheet impedance in parallel with an impressed surface current, the Norton equivalent of a source with internal resistance. Its geometry is read from the mesh: the height h is the extent of the patch along the given voltage direction and the width follows from the total area, $w = A/h$. The sheet impedance is then $Z_s = z_0 w/h$, and the port contributes $jk_0\eta_0/Z_s$ times the face mass matrix, the same expression as every other boundary condition.

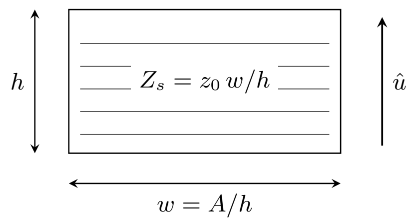


Figure 4: A lumped port. The patch of boundary triangles carries a sheet impedance and an impressed surface current. Its height is the extent of the patch along the given direction $\hat{u}$ and its width follows from the total area, so the deck states only $\hat{u}$ and the reference impedance z_0.

Derivation: port voltage without the port area

The port voltage is the mean field along the direction $\hat{u}$ times the height,

$$V = -\frac{h}{A}\int_A \mathbf{E}\cdot\hat{u}\,dA = -\frac{1}{w}\int_A \mathbf{E}\cdot\hat{u}\,dA,$$

since $A = wh$. The area cancels, so the port needs to store only its width. With port p driven, the scattering matrix follows from $S_{qp} = 2V_q/\sqrt{z_{0p}z_{0q}} - \delta_{qp}$.

One factorization serves all ports of a frequency, since only the right hand side changes. The frequencies of a sweep are independent problems and run on separate threads, each holding its own factorization, with the thread count capped by a memory budget rather than by the core count alone.

The impedance and admittance matrices follow from the scattering matrix by $Z = D(I + S)(I - S)^{-1}D$ with $D = \mathrm{diag}\sqrt{z_0}$, which needs a small dense complex solve, and $Y = Z^{-1}$, which needs a second. Reading a port as a coil gives $L = \mathrm{Im}(Z)/\omega$ and $Q = \mathrm{Im}(Z)/\mathrm{Re}(Z)$, the quantities an inductor extraction reports.

9 Results

The reference model is an edge fed microstrip patch antenna on a 1.57 mm substrate of $\varepsilon_r = 2.2$, terminated by a perfectly matched layer and meshed by Gmsh into 37676 tetrahedra and 41468 unknowns.

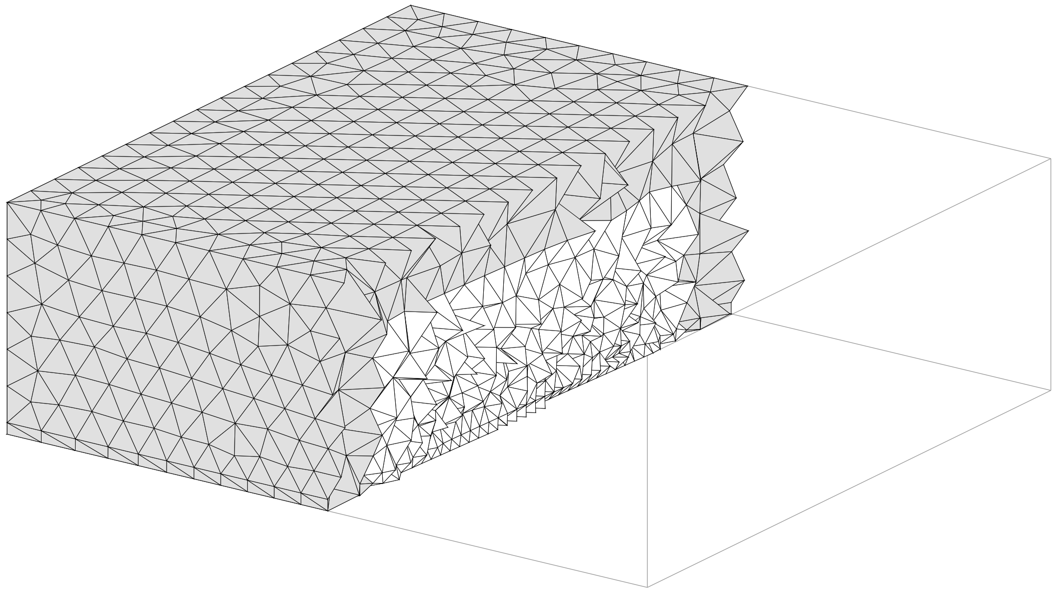


Figure 5: The mesh, cut in half at $y = 50$ mm and drawn back to front with opaque faces, inside the outline of the full domain. The thin layer at the bottom is the substrate, the block above it is the air region, and the shaded ring and cap are the perfectly matched layer. The element size follows a distance field on the patch and the feed, growing from 2 mm at the metal to 5.5 mm at the outer corners, which keeps the aspect ratio bounded through the transition out of the 1.57 mm substrate. A uniform maximum size instead produces slivers spanning the air region and moves the resonance by two percent.

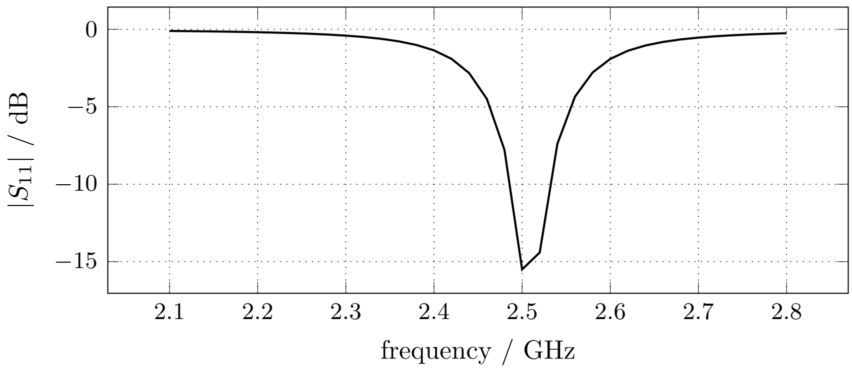


Figure 6: Reflection of the reference antenna. The resonance appears as a dip at 2.50 GHz, against a design value of 2.45 GHz. A mesh coarsened by a factor of one and a half puts it at 2.48 GHz, so the discretization error is still the largest term.

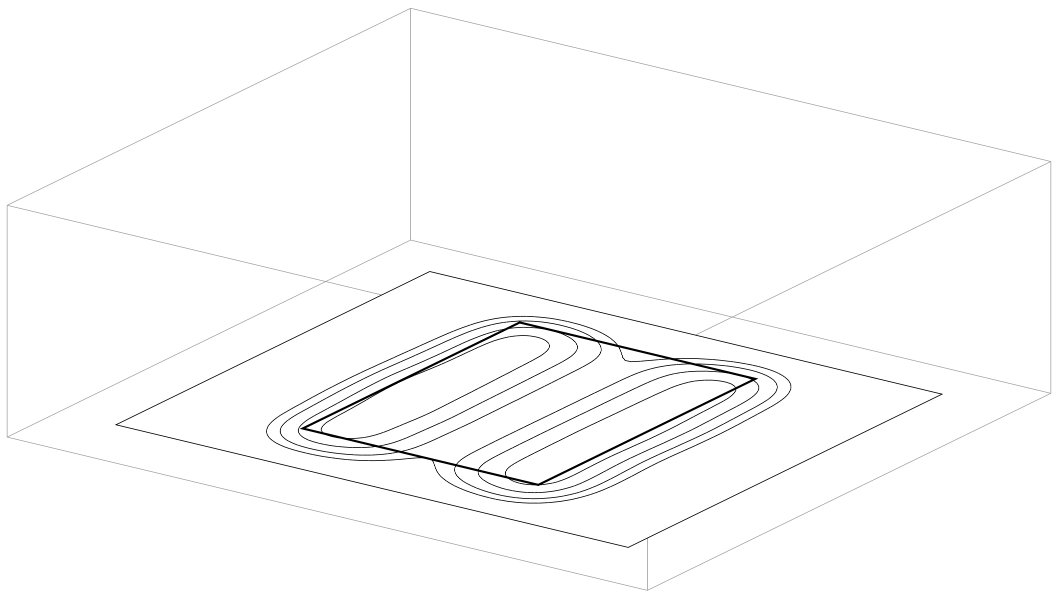


Figure 7: Contours of the electric field magnitude at the resonance, on a horizontal cut through the middle of the substrate, at −11, −7, −4 and −2 dB below the peak of the cut. The view, the scale and the outlined box are those of figure 5, so the two are the same scene: the box is the meshed domain, and the plane stands off it by the thickness of the absorbing layer. The heavy rectangle is the patch. The field is confined under the patch and forms the half wave along y that sets the resonant length, with maxima at the two radiating edges and a null across the middle. What escapes past those edges is the fringing that radiates.

stage	time	share
mesh, assembly, matrix build and ordering	35 ms	once
symbolic analysis	40 ms	once
numeric factorization	3.7 s	per frequency
triangular solve and refinement	60 ms	per frequency and port

Table 3: Runtime by stage on the reference antenna, on one core of an Apple M3. The totals are wall clock, the split between stages comes from a sampling profile of the same run. The factorization is 98 percent of the work per frequency, so the fill of the ordering sets the runtime and the assembly does not. A sweep runs its frequencies in parallel, one factorization each, which is why the memory and not the arithmetic caps the thread count.

10 Validation

Twelve integration tests run the release binary against structured meshes generated inside the test file, almost all with a closed form reference. Table 4 lists the quantitative ones.

case	quantity	nanofem	closed form
matched TEM line	$\lvert S_{21}\rvert$	1.0000	1
shorted line	$\lvert S_{11}\rvert$	1.0000	1
mismatched port	$\mathrm{Re}(S_{11})$	-0.332	$-1/3$
lossy dielectric line	$\lvert S_{21}\rvert$	0.9629	0.9630
conductive filling	$\lvert S_{21}\rvert$	0.7993	0.7984
lossy plates	$\lvert S_{21}\rvert$	0.98020	0.98019
shorted line	$\mathrm{Im}(Z_{11})$ / Ω	96.801	96.798
shorted line	L / nH	154.06	154.06
absorbing wall	$\lvert S_{11}\rvert$	0.0018	0
three cell PML	$\lvert S_{11}\rvert$	0.0293	0
PEC cavity	f_{101} / GHz	2.10	2.12

Table 4: Quantitative tests against closed form results. The cavity is fed through an aperture, which loads it and accounts for most of the difference from the closed cavity mode.

11 The cut list

feature	estimated lines	reason
second order elements	150 to 190	five times the unknowns per mesh
modal waveguide ports	200+	needs a 2D eigensolver
supernodal factorization	200+	blocked kernels for the factorization
moment matching sweeps	50+	diverges near resonances
adaptive mesh refinement	100+	needs an error estimator
dispersive materials	40	breaks the four coefficient scheme
deck validation	31	belongs to whatever writes the deck

Table 5: What the budget excluded.

Second order elements are the largest entry. They carry twenty unknowns per tetrahedron instead of six: the six Whitney functions, six curl free edge gradients, and two functions per face, the last pinned by the sorted vertex triple of each face and therefore also free of orientation handling. Their dispersion error scales as $(kh)^4$ rather than $(kh)^2$, which at six cells per wavelength is three orders of magnitude. Against that, the closed form element matrices of section 2 no longer suffice, since the integrands are no longer constant, so a quadrature rule enters; and the unknown count rises by a factor of five on a given mesh, which for a direct solver raises the memory by more than an order of magnitude. Coarsening the mesh recovers part of that, but not on the same problem geometry. Accuracy per unknown improves, the ceiling on model size falls, and the ceiling is the binding constraint here.

12 Closing

The final count is 954 of 1000 lines. The solver covers the curl-curl equation with first order Nedelec elements, perfect electric conductors, a first order absorbing boundary, perfectly matched layers, magnetic symmetry planes, dielectric loss, volume conductivity, conductor sheets with the skin effect, lumped ports with scattering, impedance, admittance and inductance output, a geometric nested dissection ordering, a complex symmetric LDL^T with equilibration and iterative refinement, a parallel frequency sweep, and field export.

Three decisions reduce code outside their own section. Sorting the mesh vertices removes orientation handling from the solver. Four frequency coefficients per matrix entry hold the assembly at one pass for an entire sweep. Storing the layer stretch separately from the permittivity lets the conduction current reuse the mass entry.